

Cato Innovation Project — Application

Applicant: Sage Arthur Jordan Clokey **Date:** June 2026

PROPOSAL (3,500 words)

The Idea: Biology Is the Strongest Case for Liberty That No One Is Making

The most powerful empirical evidence for voluntary order, limited government, and free markets is not in an economics textbook. It is in every living thing on earth — and the Cato Institute has never made this case.

Your body is 37 trillion cells. No president. No congress. No central plan. They coordinate through local signals, voluntary exchange, and distributed knowledge — exactly the way Hayek described a free market. And when one cell overrides that cooperative system — when it centralizes control and demands all resources flow to itself — we have a word for that. Cancer.

This is not a metaphor. This is peer-reviewed molecular biology. Protein-protein interaction networks show no master protein and no central regulator — proteins interact locally and global function emerges. The immune system operates as decentralized defense without a state: autonomous cells detect threats, respond immediately, and coordinate laterally without a bureaucratic approval chain. Cells that are hijacked by a pathogen would rather die than be coerced — apoptosis, programmed cell death, is a cell choosing death over submission to a hijacked program.

At the ecological scale, forests organize through mycorrhizal networks that share resources underground without central allocation. No forester designed it. Monoculture plantations — centrally planned — collapse. Diverse forests — emergent — endure for millennia.

At the genomic scale, James Shapiro's work on natural genetic engineering shows that genomes don't evolve through random character-by-character mutation — they restructure themselves by rearranging existing functional modules in new combinations, activated by stress, regulated by the cell itself. The genome operates

like a distributed ledger: no central controller, distributed consensus across cells, self-rewriting protocols, and fork events that create new species the way a hard fork creates a new chain. Life doesn't wait for a central authority to approve a mutation. It engineers its own adaptation from the bottom up.

Hayek's knowledge problem predicts that the information required to coordinate a complex system never exists in concentrated form. Biology proves this at every scale: no cell holds the full picture, yet the organism functions. Mises' calculation problem predicts that rational allocation requires prices — and biology uses price-like signals (metabolic shadow prices, hormone gradients, nutrient concentrations) to coordinate resource allocation across trillions of actors. Menger's spontaneous order predicts that functional organization arises from individual agents responding to local conditions — and this is precisely how ecosystems, immune systems, and gene regulatory networks operate.

The gap is this: Cato has world-class scholars making the economic, legal, and philosophical case for liberty. No one at Cato is making the biological case. No one is showing that the principles Hayek and Mises described are not merely economic — they are the operating principles of life itself. This is the strongest possible ground to stand on, and it is unoccupied.

The audience this unlocks is enormous. There are millions of STEM students, biologists, engineers, and doctors who intuitively understand emergent order from their training but have never connected it to political philosophy. They would never read a policy paper. But they would watch a video called "Cancer Is Central Planning" or "The Open Source Economy of the Cell." They already think in systems. They just need someone to show them that the system they study every day is the system the liberty movement has been describing in political language for decades.

This is not ideology projected onto nature. This is nature revealing the principle.

The Job: Fellow, Science and Liberty

Proposed title: Fellow, Science and Liberty

Reporting structure: I would recommend reporting to the VP of Policy or a senior fellow whose work intersects economics and institutional design. Given the cross-cutting nature of the project — spanning research, media production, and public engagement — a direct line to someone with both editorial and strategic oversight would be ideal.

Scope and responsibilities:

1. **Research and writing.** Produce original research, essays, and policy-adjacent analysis demonstrating that biological systems operate on the principles of voluntary order, distributed knowledge, and spontaneous organization. Publish through Cato's existing channels — blog posts, policy analysis papers, white papers, and the website — as well as external publications and journals at the intersection of biology and political economy.
2. **Media production.** Create video content — short-form and long-form — that translates biological evidence for liberty into visual, accessible formats. "Cancer Is Central Planning," "37 Trillion Cells, No Government," "How Does a Forest Think?" These are not academic lectures. They are Cato-branded content designed to reach audiences the Institute has never spoken to: biology students, medical students, engineers, synthetic biologists, ecologists.
3. **Public engagement.** Represent Cato at conferences, campus events, and public forums where biology, technology, and political philosophy intersect. Deliver talks at universities, appear on podcasts, and participate in panels. The "Is Nature Libertarian?" talk format — a 45-minute interdisciplinary presentation followed by open discussion — is already built and tested.
4. **Cross-disciplinary bridge-building.** Connect Cato's existing scholars with researchers in biology, bioinformatics, and complex systems who are studying voluntary order without knowing it. Facilitate conversations between economists and biologists who are describing the same principles in different languages. Build a network of STEM-aligned liberty thinkers who can contribute to Cato's mission from disciplines the Institute hasn't traditionally drawn from.
5. **Open-source tools and demonstrations.** Produce publicly available bioinformatics notebooks, data visualizations, and interactive tools that let anyone explore the evidence: protein interaction networks with no central hub, metabolic economies coordinated by price-like signals, immune systems that defend without a state. Make the data accessible and the argument self-evident.

How this fits within Cato: The Institute already has fellows and scholars covering economics, law, foreign policy, technology, education, and healthcare. This project creates a new vertical — Science and Liberty — that provides biological evidence

supporting the same principles Cato's existing scholars argue from economic and philosophical grounds. It does not duplicate existing work. It provides a new category of evidence and a new audience.

The Method

The method follows the structure of how life itself organizes: distribute the work, validate locally, let the pattern emerge.

Content production cycle (monthly): - One long-form essay or research paper connecting a specific biological system to a specific liberty principle (e.g., "The Immune System as Decentralized Defense: What Biology Teaches About Security Without a State") - Two short-form videos (3-7 minutes) visualizing the biological evidence - One public appearance — campus talk, podcast, conference panel, or Cato event

Research methodology: - Draw from peer-reviewed databases: STRING (protein-protein interactions), RegulonDB (gene regulatory networks), KEGG (metabolic pathways), NCBI (genomic data), single-cell RNA sequencing atlases - Apply Austrian economic frameworks — Hayek's knowledge problem, Mises' calculation problem, Kirznerian entrepreneurial discovery — as analytical lenses on biological data - Use bioinformatics tools (UCSC Genome Browser, NCBI Entrez, ESM-3 protein language models, AlphaFold) for original analysis and visualization - Collaborate with Cato scholars to ensure economic and philosophical rigor

Why this is consistent with Cato's principles: This project does not use biology as a metaphor for liberty. It demonstrates that liberty's core principles — voluntary exchange, distributed knowledge, spontaneous order, the failure of central planning — are the operating principles of every living system on earth. The method is empirical, the data is peer-reviewed, and the conclusions follow from the evidence. Cato's mission is to advance individual liberty through scholarship and public engagement. This project does both, from the hardest evidence available — biology itself.

The Outputs

Year 1 deliverables:

Output	Quantity	Channel
Long-form research essays	8-10	Cato blog, policy analysis, external publications
Short-form videos	15-20	Cato YouTube, social media, partner platforms
Campus/conference talks	8-10	Universities, FreedomFest, LibertyCon, SynBioBeta, academic conferences
Podcast/media appearances	10+	Cato Daily Podcast, external shows
Open-source bioinformatics notebooks	3-5	GitHub, Cato website
"Is Nature Libertarian?" event kit	1	Published for any organization to use

Year 2 deliverables:

Output	Quantity	Channel
Book manuscript	1	"The Biology of Liberty" — a Cato-published book making the full case
Long-form research	8-10	Continued production
Videos	15-20	Continued, with higher production value
University course module	1	"Biology and Political Economy" — a teachable unit any professor can adopt
Annual report	1	Measurable impact data: views, citations, audience demographics, network growth

Measurable outcomes (cumulative, end of Year 2):

Metric	Target
Total video views	250,000+
Published essays/papers	18+
Campus/conference presentations	20+
Media appearances	25+
STEM professionals connected to Cato's network	100+
Open-source tools published	8+
Book manuscript completed	Yes

About Me

I am Sage Arthur Jordan Clokey. I just graduated from the University of California, Santa Cruz with a B.S. in Bioinformatics — the study of how living systems organize, store, and process information at the molecular level.

I am also the grandson of Art Clokey, the creator of Gumby. Art took clay — dead, inert material — and breathed life into it. He gave motion to what had none. He named his studio Premavision: universal love made visible. He showed the world that the inanimate could be given a soul through patience and craft.

My father, Joe Clokey, spent eight years fighting the entertainment industry to protect Gumby's creative soul. Fox took Gumby and removed the clay, the stop-motion, the spirit — what was left was just bread. A green shape with no soul. Joe died before he could finish that fight.

The inheritance is not the character. It is the ability to bring life to inert material and to fight for its integrity. Clay to code. Story to system.

Art's medium was clay. Joe's medium was film. My medium is biology and philosophy. The through-line is the same: living things are not machines. They move themselves. They choose. They grow. And any system that denies this — whether it is an entertainment industry, a political structure, or a biological model — will be outgrown by the life it tried to contain.

What I have built:

The Living Age Trilogy. Over 100 chapters across three books. Book I — The Sagent Creed — lays the philosophical foundation for individual conscience and voluntary order. Book II — The Spiral and the Block — demonstrates how mythology and media encode the same pattern: centralized control fails, decentralized life endures. Book III — The Spiral Steward — applies Austrian economics directly to molecular biology: spontaneous order in the cell, the knowledge problem as biological, diversity as functional, the genome as a distributed ledger.

Genomic Ledger Synthesis. My capstone research at UCSC — a peer-reviewed-quality paper proposing that genome construction should follow the distributed, consensus-validated architecture of natural evolution rather than top-down central planning. The paper integrates Sc2.0 SCRaMbLE, cell-free assembly, Flux Balance Analysis shadow prices, and microbial consortium architecture. It applies Hayek's knowledge problem, Mises' calculation problem, and Kirznerian entrepreneurial discovery to genome design — and argues these principles work because biology already operates on them.

SFL State Director — California. I currently serve as the Students For Liberty State Director for California, where I’ve built a mycelium-model campus network, created the “Is Nature Libertarian?” Socratic event format, and connected biology students, engineers, and philosophy students across California campuses to the principles of voluntary order. This summer I’m attending FreedomFest, Approaching Humanity, FEE Summer Campus, and YALCON — all in July.

SynBioBeta connections. At SynBioBeta 2026, I connected with Drew Endy — professor of bioengineering at Stanford and fellow at the Hoover Institution — on how to do synthetic biology in ways aligned with voluntary principles. The overlap between decentralized biology and libertarian philosophy is natural. These people already think in emergent systems.

I care less about credentials than about whether the work is true and whether it reaches the people who need to hear it. The biology of liberty is true. Cato is where it needs to be heard.

Mission Alignment

Individual liberty. Biology demonstrates that every living system operates through individual agents making autonomous decisions based on local knowledge. No cell is commanded. No protein is conscripted. Voluntary action is the foundation of all biological function. When individual autonomy is overridden — by a virus, a pathogen, or a cancerous cell — the system fails. Liberty is not a political preference. It is how life works.

Limited government. The biological analog of unlimited government is cancer: a cell that overrides the cooperative system, centralizes resource allocation, and demands that all activity serve its growth. Every centralized biological control mechanism — from viral hijacking to autoimmune overreaction — produces disease. Limited, distributed governance is the only model that sustains life at any scale.

Free markets. Biological systems coordinate through price-like signals: metabolic concentrations, hormone gradients, nutrient availability. These signals encode distributed knowledge that no central planner can hold. Hayek’s price system is not a metaphor for biology — biology is the original implementation. Cells trade molecular goods through voluntary exchange. Organisms specialize based on comparative advantage. Ecosystems allocate resources through decentralized negotiation. The free market is four billion years old.

Peace. The immune system is the biological model of defense without empire. It protects the organism without conquering other organisms. It distinguishes self from non-self without aggression toward everything foreign. It maintains tolerance — immunological peace — as its default state, and responds with force only when the boundary is violated. Defense, not offense. Protection, not conquest. This is peace as biology practices it.

What You Would Want to Know

“Is this real science or just metaphor?” It is real science. Every claim is grounded in peer-reviewed biology: protein-protein interaction networks (STRING database), gene regulatory networks (RegulonDB), metabolic pathway analysis (KEGG, Flux Balance Analysis), genomic architecture (NCBI, UCSC Genome Browser), and immunology. I hold a B.S. in Bioinformatics from UC Santa Cruz. The tools I use — ESM-3 protein language models, AlphaFold, single-cell RNA sequencing analysis — are the standard tools of the field. The Austrian economic frameworks I apply are analytical lenses, not metaphors. The data stands on its own.

“Can you actually produce at this volume?” I wrote over 100 trilogy chapters, a capstone research paper, a state director strategy, event templates, tabling playbooks, video scripts, and pitch documents — all while finishing my degree. I use AI (Claude) as a core workflow tool for research, drafting, and analysis. The production system is already built.

“Will anyone watch this?” The “Cancer Is Central Planning” framing stops people in their tracks. I have tested it — at campus events, at tabling sessions, in one-on-one conversations. Biology students, medical students, engineers, philosophy students — they all respond the same way: “I’ve never heard it put that way.” That reaction is the signal. The audience exists. They have never been spoken to in their own language by the liberty movement. They are waiting.

“Why Cato and not a university or a think tank focused on science?” Because the insight is not scientific — it is political-economic. A biology department can describe how cells coordinate. Only Cato can explain why that coordination is the same principle Hayek described, why it matters for policy, and why it should change how we think about governance. The institutional credibility of Cato turns a biological observation into a political-economic argument that demands engagement. And Cato’s platform — the website, the podcast, the events, the media presence — gives this work an audience that no university lab can match.

“Will this appeal to Cato’s existing audience?” Yes — and it will expand it. Cato’s existing audience already understands Hayek and Mises. Showing them that biology confirms what they believe gives them new ammunition: harder evidence, fresher language, and an argument that cannot be dismissed as ideology because it comes from nature itself. But the real value is the new audience — the millions of STEM-trained minds who have never engaged with Cato because they didn’t know Cato was describing the systems they study every day.

“Are you willing to relocate to DC?” Yes.

PITCH SIMPLIFICATION (50 words)

Biology is the strongest case for liberty no one is making. Your body is 37 trillion cells with no government — and when one centralizes control, we call it cancer. I want to build Cato’s Science and Liberty program: research, video, and outreach proving voluntary order is how life works.

BUDGET BREAKDOWN

Category	Year 1	Year 2	Notes
Salary	\$100,000	\$100,000	Per program terms
Benefits	Standard	Standard	Per Cato employee package
Video production (equipment, editing software, graphics)	\$5,000	\$3,000	Camera, mic, lighting, Adobe suite, motion graphics. Year 2 lower — equipment purchased in Year 1
Travel (campus talks, conferences, research)	\$8,000	\$8,000	8-10 events/year across universities and conferences (FreedomFest, LibertyCon, SynBioBeta, ISFLC, academic)
Data and software subscriptions	\$2,000	\$2,000	Bioinformatics databases, cloud compute for protein analysis, design tools
Book production (editing, design, printing)	\$0	\$5,000	Year 2 — “The Biology of Liberty” manuscript editing and publication
Event hosting and materials	\$2,000	\$2,000	Campus event materials, printed handouts, event promotion
Research materials and books	\$1,000	\$500	Primary sources, reference texts, journal access beyond Cato’s existing subscriptions
Total project budget (beyond salary)	\$18,000	\$20,500	
Total annual cost	\$118,000	\$120,500	Salary + project budget

SUPPORTING MATERIALS (to attach)

1. [statement_fellowship_application.pdf](#) — Full Living Age project proposal (demonstrates scope and depth of existing work)
2. [genomic_ledger_synthesis.pdf](#) — Capstone research paper: Genomic Ledger Synthesis (demonstrates scientific credibility)
3. Stossel Fellowship 1-minute video (if recorded) — as multimedia URL

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